THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET C

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The nth term of the AP $7, 11, 15, \ldots$ is: 1

(a) $4n + 3$ **(b)** $7n + 4$ **(c)** $3n + 4$ **(d)** $4n + 7$

2. For the sum of the first 20 natural numbers, Student A gets 210 and Student B gets 200. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

3. Which of the following is an AP? 1

(a) $1, 1, 2, 3, \ldots$ **(b)** $-10, -6, -2, 2, \ldots$ **(c)** $2, 4, 8, 16, \ldots$ **(d)** $1, 3, 9, 27, \ldots$

4. The sum of the first n even natural numbers is: 1

(a) $n(n+1)$ **(b)** n^2 **(c)** $2n^2$ **(d)** $n(n-1)$

5. $\triangle ABC \sim \triangle DEF$ with $AB = 3$ cm, $DE = 6$ cm and $BC = 4$ cm. Then EF is: 1

(a) 8 cm **(b)** 2 cm **(c)** 6 cm **(d)** 12 cm

6. In $\triangle ABC$, $DE \parallel BC$, $\frac{AD}{AB} = \frac{2}{5}$ and $DE = 4$ cm. Then BC is: 1

(a) 10 cm **(b)** 12 cm **(c)** 8 cm **(d)** 6 cm

7. If $\angle A = \angle P$ and $\angle B = \angle Q$ in $\triangle ABC$ and $\triangle PQR$, the triangles are similar by: 1

(a) RHS **(b)** SAS **(c)** SSS **(d)** AA

8. All squares are: 1

(a) similar **(b)** congruent **(c)** neither **(d)** equal in area

9. The mid-point of the segment joining $(2, -3)$ and $(6, 5)$ is: 1

(a) $(8, 2)$ **(b)** $(4, 1)$ **(c)** $(4, -1)$ **(d)** $(2, 4)$

10. If the distance between $(4, p)$ and $(1, 0)$ is 5, then p is: 1

(a) −4 only **(b)** 0 **(c)** ± 4 **(d)** 4 only

11. The point $(-3, 0)$ lies: 1

(a) on the negative x-axis **(b)** in Quadrant III

(c) in Quadrant II **(d)** on the y-axis

12. The point $P(2, 4)$ divides the join of A(0, 2) and B(3, 5) in the ratio: 1

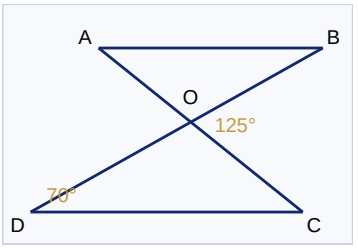
(a) 1 : 2 **(b)** 2 : 1 **(c)** 3 : 2 **(d)** 1 : 1

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. A student claims that -150 is a term of the AP $11, 8, 5, 2, \ldots$ Justify or refute. 2

14. In the figure, $\triangle ODC \sim \triangle OBA$, $\angle BOC = 125°$ and $\angle CDO = 70°$. Find $\angle DOC$, $\angle DCO$ and $\angle OAB$. **2**



15. In $\triangle ABC$, $DE \parallel BC$ where D is on AB and E is on AC. If $AD = 6$ cm, $AE = 8$ cm and $EC = 4$ cm, find DB. **2**

16. A student claims that A(3, 0), B(6, 4) and C(−1, 3) are the vertices of a right isosceles triangle. Verify. **2**

17. A student claims that the points A(1, 5), B(2, 3) and C(−2, −11) are collinear. Justify or refute. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. A contract on a construction job specifies a penalty for delay: ₹200 for the first day, ₹250 for the second day, ₹300 for the third day, and so on. How much money does the contractor pay as penalty if he delays the work by 30 days? **3**

19. In $\triangle ABC$, D and E are points on AB and AC such that $AD = AE$ and $AB = AC$. Show that $\triangle ADE \sim \triangle ABC$. **3**

20. ABCD is a trapezium with $AB \parallel DC$. E and F are points on AD and BC such that $EF \parallel AB$. Show that $\frac{AE}{ED} = \frac{BF}{FC}$. **3**

21. If the point (x, y) is equidistant from $(7, 1)$ and $(3, 5)$, show that $x - y = 2$. **3**

22. If A(1, 2), B(4, y), C(x, 6) and D(3, 5) are the vertices of a parallelogram taken in order, find x and y. **3**

OR

Find the coordinates of the points of trisection of the segment joining A(2, −2) and B(−7, 4). **3**

23. In an AP, $a = 5$, $d = 3$ and $a_n = 50$. Find n and S_n. **3**

OR

The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. **5**

OR

Prove that if a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. **5**

25. **Case study:** A port is at P(0, 0); two ships are at S_1(6, 8) and S_2(−8, 6) (units in km). **5**

(i) Find their distances from the port.

(ii) Find the distance between the ships.

(iii) A rescue boat waits midway between the ships. Find its position.

(iv) Show that $\angle S_1PS_2 = 90°$.

OR

Find the coordinates of the points of trisection of the segment joining $(4, -1)$ and $(-2, -3)$. Verify, using the distance formula, that the first trisection point is equidistant from $(4, -1)$ and the second trisection point. **5**

END OF QUESTION PAPER — **Total: 50 Marks**